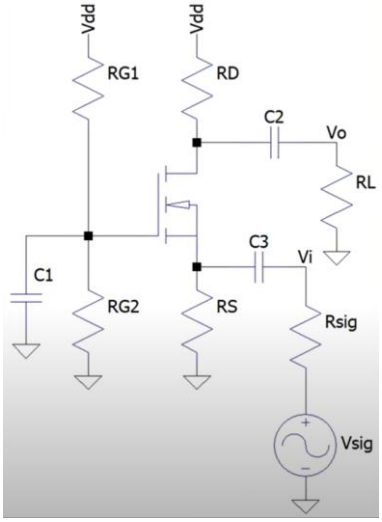
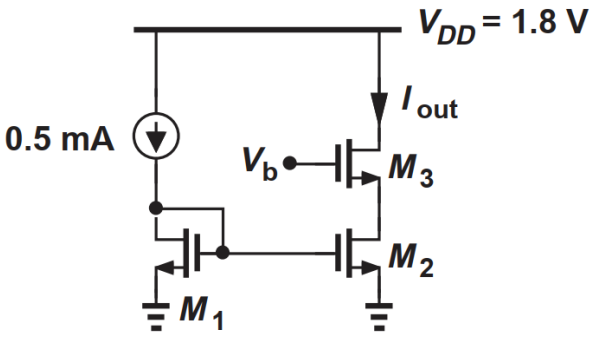
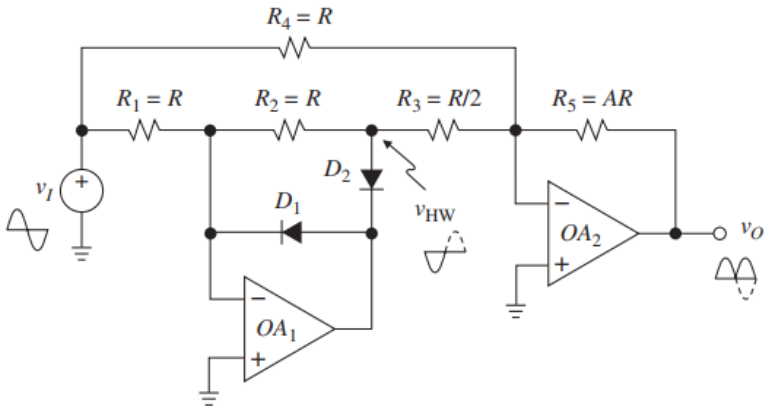
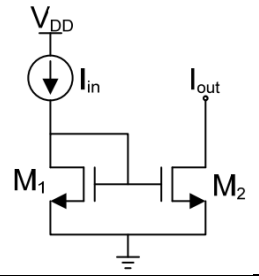
Assignment
III Semester B.Tech VLSI
Subject: Analog circuits (ECE2126) (Odd semester- 2025)

Instructions

1. **Answer all the questions.**
2. **Use either LTspice or Electric software for simulation. You may use 180nm technology node.**
3. **Provide hand calculation wherever it is necessary depending on the question**
4. **Missing data may be suitably assumed and clearly mention the same**

1	Design and simulate a common source amplifier with diode connected PMOS as load and obtain the following 1. A gain (Av) of ____ (last three digits of your Reg. No). 2. Obtain the DC bias voltage needed to operate the transistor in saturation point. Further, explain the method that you adopted for the same 3. Plot the DC and transient characteristics of the CS amplifier Note : Consider Vdd = 5V, W= 5µM. Vin = 5×10^{-3} sin(20000πt)
2	a. Explain the importance of source follower with ideal current source. b. Design and simulate a source follower with ideal current source for Vdd = 5V, W= 5µM. Vin = 5×10^{-3} sin(20000πt). c. Plot the transient waveform and comment on the i) gain ii) input impedance and iii) output impedance
3	A common gate amplifier needs to amplify an input of Vin = 1×10^{-3} sin(20000πt) which is been loaded with a 22k Ω load resistor(refer figure). Assume a 100µA current flowing in the bias resistors, VDD = 12V, Zin = 50 Ω. a. Design: RD, RS, RG1 and RG2 b. Calculate: Zin, Zout, and AV. Assume Rsig = 50 Ω c. Simulate the circuit and plot transient waveform for Vin and Vout 
4	Figure depicts a cascode current source whose value is defined by the mirror arrangement, M1-M2. Assume W/L = 5 µm/ 0.18 µm for M1-M3. a. Select the value of Vb so that Iout is precisely equal to 0.5 mA.

	b. Determine the change in Iout if Vb varies by ±100 mV. Explain the cause of this change. c. Using hand calculation, determine the output impedance of the cascode section. 
5	Suitably modify the FWR of Fig. shown, so that, when fed with a triangular wave of ±5V peak values, it gives a triangular wave of ±5V peak values, but twice the frequency. Assume ±10V regulated power supplies. 
6	Simulate the following circuit (with suitable Rout) and plot I_{in} Versus I_{out} when W/L ratio of a) M1=M2 b) M1=2M2 c) M1=M2/2. Assume Vdd = 1.8V, W/L = 5 µm/ 0.18 µm, I_{in} = 0.5mA 
	Reference: 1. https://www.youtube.com/watch?v=49naoDY34nU 2. https://github.com/ksooryakrishna1/Using-TSMC-Model-Files-350nm-250nm-180nm-any-technology-model-file-in-LTspice-

3. https://www.staticfreesoft.com/index.html
4. https://www.youtube.com/@vlsiacademictutorialsbypra5207
5. https://www.youtube.com/@EzElectronics/videos